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5 / 5 submitted

Q1. Harshad number

ANSWER

```
def is_harshad(n):
    digit_sum = sum(int(d) for d in str(n))
    return n % digit_sum == 0

N = int(input("Enter a number: "))

if is_harshad(N):
    print(f"{N} is a Harshad number.")
else:
    print(f"{N} is not a Harshad number.")
print("\nHarshad numbers between 1 and 500:")
harshad_list = [n for n in range(1, 501) if is_harshad(n)]
print(harshad_list)
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = b
    b = a % b
    a = temp
print("GCD is", a)
```

INPUT

```
48
18
```

OUTPUT

```
Enter first: Enter second: GCD is 6
```

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
        seen.add(num)
    return pairs
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data, t)
print("Pairs:", result)
```

Q4. Bitwise Operators

ANSWER

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

CODE

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1
    while low ≤ high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if arr[mid] < target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1
    return -1

data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))
result = rotated_binary_search(data, t)
print("Found at:", result)
```

INPUT

2

OUTPUT

Target: Found at: 6